Please submit your solutions to the following problems on Gradescope by **6pm** on the due date. Collaboration is encouraged, however, you must write up your solutions individually.

**1) A one-dimensional oscillator.** Consider the linear differential operator

$$Au = u'' + 4\pi^2 u$$

on the space $V = C^\infty_{\text{per}}([0,1])$ of smooth complex-valued periodic functions with inner product

$$\langle u, v \rangle = \int_0^1 u(x)\overline{v(x)}\, dx.$$

Following parts (a)-(d), characterize the smooth periodic solutions to $Au = f$ with $u \in V$.

(a) Solve $Au = \lambda u$ for $u \in V$, i.e., compute the smooth, periodic eigenfunctions of $A$ and the associated eigenvalues.

(b) Calculate the adjoint $A^*$ of $A$ (using the inner product $\langle \cdot, \cdot \rangle$) and compute orthonormal bases for the four fundamental subspaces: $\mathcal{N}(A)$, $\mathcal{R}(A)$, $\mathcal{N}(A^*)$, and $\mathcal{R}(A^*)$.

(c) Derive necessary and sufficient conditions on $f \in V$ for $Au = f$ to have a solution in $V$. Then, consider the following two forcings:

(i) For $f(x) = \cos(4\pi x)$, find all periodic solutions and explicitly identify any free parameters in your solution.

(ii) For $f(x) = \cos(2\pi x)$, determine whether a periodic solution exists. Find one non-periodic solution of the differential equation, verify it by substitution, and identify which periodic boundary condition it fails.

(d) In place of $A$, consider the linear differential operator associated with a damped oscillator,

$$A_\gamma u = u'' + 2\gamma u' + 4\pi^2 u, \qquad \gamma > 0,$$

on the same space $V$ of smooth, periodic functions. Calculate the eigenvalues of $A_\gamma$ and, for $f(x) = \cos(2\pi x)$, calculate a periodic solution to $A_\gamma u = f$. Is it unique? How does the amplitude $\max_{0 \le x \le 1} |u(x)|$ depend on $\gamma$, and what happens as $\gamma \to 0^+$?

**2) A Sturm–Liouville problem.** Consider the Sturm–Liouville eigenvalue problem

$$u''(x) - q(x)u(x) = \lambda u(x), \qquad u(0) = u(1) = 0.$$

Here, $q(x)$ is a continuous, real-valued, non-negative function and we are looking for eigenfunctions $u(x)$ and associated eigenvalues $\lambda$ that solve this second-order differential equation.

(a) Show that any eigenvalue of the Sturm–Liouville problem must be strictly negative.

**Hint:** Integrate both sides against a carefully selected test function and integrate by parts to show that $-\lambda$ is the integral of a strictly positive function.

(b) Show that any two eigenfunctions with distinct eigenvalues must be orthogonal.

**Hint:** Use the fact that the differential operator $Lu = u'' - qu$ is self-adjoint.

(c) Let $v_-$ satisfy $v_-'' - qv_- = 0$ subject to $v_-(0) = 0$ and $v_-'(0) = 1$, and $v_+$ satisfy $v_+'' - qv_+ = 0$ subject to $v_+(1) = 0$ and $v_+'(1) = 1$. Show that the "Wronskian" $w = v_+'(x)v_-(x) - v_-'(x)v_+(x)$ of these two solutions is a nonzero constant and define the integral operator $[Tf](x) = \int_0^1 k(x,y)f(y)\,dy$ with

$$k(x,y) = \begin{cases} w^{-1}v_-(x)v_+(y), & 0 \le x \le y \le 1, \\ w^{-1}v_+(x)v_-(y), & 0 \le y \le x \le 1. \end{cases}$$

Verify that $T$ is a self-adjoint Hilbert–Schmidt operator.

(d) Let $T$ be the integral operator in (c). Show that if $f$ is continuous on $[0,1]$, then

$$[Tf]''(x) - q(x)[Tf](x) = f(x),$$

i.e., that $T$ is a bounded inverse of the differential operator $[Lu](x) = u''(x) - q(x)u(x)$.

(e) Use your work in (a)-(d) to argue that each eigenfunction of $T$ is an eigenvector of $L$ and explain why this implies that the orthogonal eigenfunctions of $L$ span $L^2([0,1])$.